

WEEK 7 – POWER BI REPORT

STUDENT PERFORMANCE ANALYSIS – DATA PREPARATION

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1. Overview

The Student Performance dataset was processed in Microsoft Power BI for further academic analysis. The raw dataset was improved by carrying out data cleaning, transformation, calculated field creation, and restructuring activities.

The processed dataset provides a suitable base for developing the Student Performance Dashboard.

2. Aim of the Project

The primary objective of this project is to convert the original student examination data into a properly organized dataset that can be used for analysis.

Key Goals

- Load the student dataset into Power BI.
 - Inspect the structure and overall quality of the dataset.
 - Identify and remove unwanted or inconsistent information.
 - Modify categorical and numerical fields where required.
 - Derive total and average marks.
 - Divide students into performance categories.
 - Assign Pass or Fail status.
 - Rearrange the subject-level score information.
 - Develop DAX measures for further analysis.
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3. Dataset Description

The dataset provides details about students along with their marks obtained in different examinations.

Major Fields

Category	Fields
Student Details	Gender, Race/Ethnicity
Background	Parental Education, Lunch
Preparation	Test Preparation Course

Category	Fields
Academic Scores	Math, Reading, Writing

These attributes make it possible to examine student performance using different categories and perspectives.

Student ID	gender	race/ethnicity	parental level of education	lunch	test preparation course
1	Female	Group B	Bachelor's Degree	Standard	None
2	Female	Group C	Some College	Standard	Completed
3	Female	Group B	Master's Degree	Standard	None
4	Male	Group A	Associate's Degree	Free/Reduced	None
5	Male	Group C	Some College	Standard	None
6	Female	Group B	Associate's Degree	Standard	None
7	Female	Group B	Some College	Standard	Completed
8	Male	Group B	Some College	Free/Reduced	None
9	Male	Group D	High School	Free/Reduced	Completed
10	Female	Group B	High School	Free/Reduced	None
11	Male	Group C	Associate's Degree	Standard	None
12	Male	Group D	Associate's Degree	Standard	None
13	Female	Group B	High School	Standard	None
14	Male	Group A	Some College	Standard	Completed
15	Female	Group A	Master's Degree	Standard	None
16	Female	Group C	Some High School	Standard	None
17	Male	Group C	High School	Standard	None
18	Female	Group B	Some High School	Free/Reduced	None
19	Male	Group C	Master's Degree	Free/Reduced	Completed
20	Female	Group C	Associate's Degree	Free/Reduced	None
21	Male	Group D	High School	Standard	None
22	Female	Group B	Some College	Free/Reduced	Completed

4. Power Query Data Preparation

After importing the dataset into Power BI, it was opened in the Power Query Editor for preprocessing.

The dataset was inspected for the following issues:

- Incorrect column data types
- Missing information
- Repeated records
- Invalid entries
- Non-uniform category values
- Columns that were not required

The identified problems were addressed before moving forward with the remaining data preparation steps.

5. Data Cleaning Process

Several preprocessing operations were performed to improve the quality and consistency of the dataset.

Duplicate Removal

Repeated records were identified and eliminated wherever necessary.

Missing Value Handling

The dataset was checked for missing or invalid entries, and appropriate handling was performed.

Data Type Correction

Suitable data types were assigned to the columns to ensure accurate calculations and processing.

Category Standardization

Categorical fields were examined and made consistent for reliable analysis. WEEK 7 merlyn

6. Student Score Calculation

Additional calculated fields were introduced to obtain a summary of each student's examination performance.

Total Marks

Total Score = Math Score + Reading Score + Writing Score

Average Marks

Average Score = Total Score ÷ 3

These calculated values help represent the student's overall academic performance in a simple and measurable form.

Custom Column

Add a column that is computed from the other columns.

New column name
Total Marks

Custom column formula
= [math score]+[reading score]+[writing score]

Available columns
lunch
test preparation course
math score
reading score
writing score
Total Marks
Average

<< Insert

Learn about Power Query formulas

✓ No syntax errors have been detected.

OK Cancel

Custom Column

Add a column that is computed from the other columns.

New column name
Average

Custom column formula
= [Total Marks]/3

Available columns
lunch
test preparation course
math score
reading score
writing score
Total Marks
Average

<< Insert

Learn about Power Query formulas

✓ No syntax errors have been detected.

OK Cancel

7. Performance Categorization

Conditional logic was applied to place students into different performance groups.

Grade Performance Level

- A High Performance
- B Good Performance
- C Average Performance
- F Low Performance

This grouping can be used to understand how students are distributed across different achievement levels.

8. Pass and Fail Analysis

A conditional column was introduced to classify students into two categories: **Pass** and **Fail**.

This classification can later be used to determine the overall pass percentage and analyze differences in performance among various student groups. WEEK 7 merlyn

Table.AddColumn(# Added Conditional Column , result , each if (# Average) />= 40 then Pass else Fail)

Add Conditional Column

Add a conditional column that is computed from the other columns or values.

New column name
Result

	Column Name	Operator	Value		Output
If	Average	is greater than or...	40	Then	Pass

Add Clause

Else
Fail

OK Cancel

9. Subject Data Transformation

The dataset originally contained separate score columns for:

- Math Score
- Reading Score
- Writing Score

These three columns were converted into a structured format using the **Unpivot** operation in Power Query.

After transformation, the data was represented using:

New Field	Description
Subject	Examination subject name
Score	Marks obtained by the student

This arrangement makes it easier to perform comparisons and analysis based on individual subjects.

Student ID	Subject	Score
1	math score	72
2	reading score	72
3	writing score	74
4	math score	69
5	reading score	90
6	writing score	88
7	math score	90
8	reading score	95
9	writing score	93
10	math score	47
11	reading score	57
12	writing score	44
13	math score	76
14	reading score	78
15	writing score	75
16	math score	71
17	reading score	83
18	writing score	78
19	math score	88
20	reading score	95
21	writing score	92
22	math score	40
23	reading score	43
24	writing score	29

10. Data Model Preparation

Once the required transformations were completed, the dataset was reviewed before being used in Power BI.

Preparation Steps

- Removed fields that were not required.
- Examined the transformed table.
- Checked the available keys and relationships.
- Confirmed the structure of the final dataset.
- Organized the table for visualization purposes. WEEK 7 merlyn

11. DAX Measures

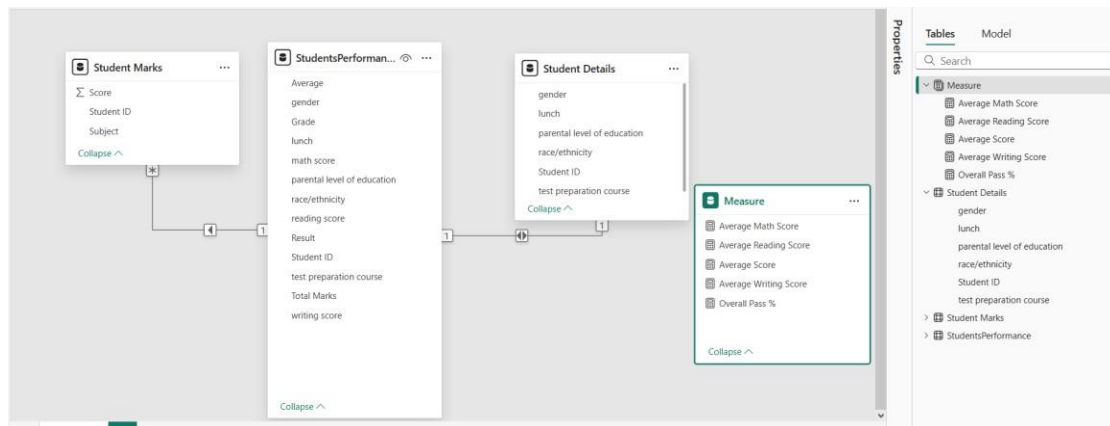
Basic DAX measures were developed to provide summarized academic performance information for the dashboard.

Measures Created

- Average Math Score
- Average Reading Score

- Average Writing Score
- Overall Average Score
- Overall Pass %

These measures help summarize important performance indicators for further analysis.



12. Data Verification

The completed dataset was reviewed to confirm that the applied transformations and calculations were functioning correctly.

The following elements were validated:

- Total Score calculation
- Average Score calculation
- Performance Grade
- Pass/Fail classification
- Subject-wise transformation
- DAX measure outputs

After completing these checks, the transformed dataset was considered suitable for visualization.

13. Expected Analysis

The prepared dataset can be used to investigate the following areas:

- Overall academic performance of students
- Performance across individual subjects
- Differences in results among student groups
- Influence of test preparation on scores
- Connection between parental education and student performance
- Pass and Fail percentages
- Distribution of performance grades WEEK 7 merlyn

14. Result

The original Student Performance dataset was successfully processed and converted into a structured format suitable for analysis.

Final Dataset Includes:

Cleaned Data + Calculated Scores + Performance Grade + Pass/Fail Status + Unpivoted Subject Data + DAX Measures

The resulting dataset provides the necessary foundation for developing the Student Performance Dashboard.

15. Conclusion

The Week 7 activity provided practical understanding of how data cleaning and transformation can be performed in Power BI. The raw examination dataset was examined, cleaned, transformed, and organized for analytical use.

Calculated columns and DAX measures were created to summarize student performance. The completed dataset can now be used for dashboard creation and visualization in the following stage. WEEK 7 merlyn

Data Preparation Flow

Raw Dataset → Data Inspection → Data Cleaning → Transformation → Calculated Fields → Unpivot → DAX Measures → Validation → Dashboard Ready

